AI-POWERED REAL ESTATE

Decision Intelligence Platform

Business Requirements Document

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DOCUMENT COVERS

Executive Summary · Business Problem · AS-IS / TO-BE · Stakeholders · Scope · Business Process · Requirements · Constraints · Success Criteria

TABLE OF CONTENTS

Document Index

01 Executive Summary

Project overview and strategic context

The AI-Powered Real Estate Decision Intelligence Platform aims to transform traditional real estate reporting into an intelligent decision-support ecosystem. The organisation currently collects data across properties, sales, leads, marketing campaigns, agents, and customer feedback. However, this data remains fragmented and requires significant manual effort to analyse and convert into actionable business insights.

This initiative will establish a centralised data platform using PostgreSQL, automate data processing through n8n workflows, generate business intelligence through SQL analytics, and leverage Large Language Models (LLMs) to provide AI-generated market insights and investment recommendations. The final solution will equip management with real-time dashboards, automated reports, and AI-assisted decision-making capabilities.

Strategic Objectives

CENTRALISE	AUTOMATE	ANALYSE	PREDICT
All business data in PostgreSQL warehouse	End-to-end workflows via n8n & Python	SQL + Power BI KPI dashboards	LLM-powered market intelligence & AI recs

02 Business Problem Statement

Challenges driving this initiative

The organisation currently lacks an integrated platform capable of transforming operational real estate data into actionable business intelligence. Property performance, lead generation, sales activity, marketing effectiveness, and customer feedback are all analysed manually through spreadsheets and disconnected reports.

As data volumes increase, manual analysis becomes inefficient, error-prone, and unable to provide timely decision support. Management requires a solution capable of automating data processing, generating insights, identifying opportunities, and delivering strategic recommendations.

FRAGMENTED DATA SOURCES
Property, sales, leads, marketing, and feedback data exist in silos with no central repository.

MANUAL REPORTING BURDEN
Analysts spend excessive time compiling reports rather than interpreting insights.

DELAYED DECISION SUPPORT
Insights are delivered too slowly to enable timely operational or strategic responses.

NO PREDICTIVE CAPABILITY

Current tooling provides only backward-looking analysis with no forecasting or AI intelligence.

03 Current State Analysis

AS-IS — How the business operates today

The following describes the current state of business operations prior to implementing the Decision Intelligence Platform. Understanding the AS-IS environment is critical for identifying pain points, defining transition requirements, and measuring future improvement.

#	Current State Characteristic	Impact
1	Manual data analysis dominates all reporting cycles	High effort, low speed
2	Property data maintained in disconnected spreadsheets	Data quality risk
3	Sales and lead performance reviewed via separate reports	No unified view
4	Marketing effectiveness difficult to measure in real time	Budget misallocation
5	Business decisions reliant on manual historical interpretation	Delayed responses
6	No centralised intelligence platform currently exists	Zero AI capability

04 Future State Analysis

TO-BE — The target operating model

The following describes the desired future state upon full delivery of the AI-Powered Real Estate Decision Intelligence Platform. Each capability represents a direct resolution to the current state gaps identified above.

#	Future Capability	Business Outcome
1	Centralised PostgreSQL data warehouse	Single source of truth for all data
2	Automated ingestion, validation & processing (n8n)	Zero manual ETL effort
3	Auto-generated business metrics and KPIs	Real-time performance visibility
4	LLM-powered trend analysis and recommendations	AI-assisted strategic decisions
5	Power BI executive dashboards with live data	Instant C-suite reporting
6	RAG knowledge layer for contextual AI queries	Natural language data access

05 Stakeholders

Project participants, owners and decision-makers

The following stakeholders have been identified as key participants, decision-makers, and impacted parties for this project. Stakeholder engagement and sign-off is required at each project milestone gate.

C

Chief Executive Officer
EXECUTIVE LEADERSHIP

Project sponsor. Final approver of strategic direction and investment.

H

Head of Sales
SALES DIVISION

Primary consumer of sales dashboards. Defines agent KPI requirements.

M

Marketing Director
MARKETING DIVISION

Owns campaign ROI requirements. Approves marketing analytics scope.

H

Head of Operations
OPERATIONS

Responsible for process definitions, AS-IS documentation and UAT sign-off.

D

Data Engineering Lead
TECHNOLOGY TEAM

Technical delivery owner. Responsible for PostgreSQL, ETL and AI integration.

B

Business Analyst
PMO / ANALYSIS

Documents requirements, facilitates workshops and manages traceability.

F

Finance Controller
FINANCE DIVISION

Reviews budget and validates financial KPIs within dashboards.

I

IT / Infrastructure Manager
TECHNOLOGY TEAM

Manages infrastructure, connectivity and security compliance.

06 Project Scope

In-scope deliverables and out-of-scope exclusions

The following scope boundaries define what this project will and will not deliver. Any items not listed as in-scope are considered out-of-scope by default and require a formal change request to include. Scope changes must be approved by the project sponsor.

IN SCOPE	OUT OF SCOPE
PostgreSQL centralised data warehouse design and build	CRM system development or replacement
Python / Pandas ETL pipeline for data cleaning	Property listing portal or customer-facing website
n8n workflow automation for ingestion and alerts	Mobile application development
SQL analytical views, KPI calculations and metrics	Third-party API integrations beyond LLM services
LLM integration for AI-generated insights and recommendations	Financial accounting or ERP system changes
Power BI dashboard suite (Executive, Sales, Marketing, CX)	Legacy data migration beyond defined source files
Retrieval-Augmented Generation (RAG) knowledge layer	Ongoing infrastructure management post-go-live
Data quality validation framework	Staff recruitment or HR processes
User acceptance testing (UAT) and stakeholder sign-off	Legal or compliance advisory services
Technical documentation and handover	External market data subscriptions

Key Deliverables

ID	Deliverable	Description	Phase
D01	PostgreSQL Data Warehouse	Centralised relational database for all business data	Phase 1
D02	Python ETL Pipeline	Automated data cleaning and transformation scripts	Phase 1
D03	n8n Workflow Automation	Scheduled ingestion, validation and KPI alert workflows	Phase 2
D04	SQL Analytics Layer	Views, metrics and KPI calculation models	Phase 2
D05	LLM Intelligence Engine	AI-powered market insights and investment recommendations	Phase 3
D06	Power BI Dashboard Suite	Four interactive executive dashboards	Phase 3
D07	RAG Knowledge Layer	Contextual NL query engine over business data	Phase 4
D08	Documentation & Handover Pack	Technical docs, user guides and UAT sign-off records	Phase 4

07 Business Process

End-to-end data and intelligence workflow

AS-IS Business Process

Today, all data collection, cleaning, analysis and reporting is performed manually by business analysts and operations staff. The process is time-consuming, error-prone, and unable to scale as data volumes grow.

STEP 01	STEP 02	STEP 03	STEP 04	STEP 05
Collect	**Clean**	**Analyse**	**Report**	**Decide**
Manual data collection from Excel/CSV files	Manual spreadsheet cleaning and deduplication	Ad-hoc manual calculations and pivot	Static PDF/Excel reports distributed by email	Management reviews reports in meetings

TO-BE Business Process

The future automated process eliminates manual steps, enforces data quality at each stage, and delivers real-time intelligence directly to decision-makers through Power BI and AI-generated insights — reducing time-to-insight from days to seconds.

STEP 01	STEP 02	STEP 03	STEP 04	STEP 05
Ingest	**Store**	**Automate**	**Analyse**	**Visualise**
Python ETL pulls & cleans source files automatically	Validated data loaded into PostgreSQL warehouse	n8n triggers KPI alerts and scheduled analytics	LLM generates insights and investment	Power BI dashboards provide real-time executive

Data Flow Architecture

Layer	Component	Technology	Purpose
Source	Excel / CSV Files	Flat Files	Raw operational data input
Processing	Data Cleaning & Validation	Python/Pandas	ETL transformation and quality
Staging	Clean CSV Layer	CSV Files	Validated intermediate output
Storage	Data Warehouse	PostgreSQL	Central analytical data store
Automation	Workflow Engine	n8n	Orchestration and KPI alerts
Intelligence	AI Query Engine	Gemini LLM	Natural language analytics
Reporting	Executive Dashboards	Power BI	Business intelligence output

08 Proposed Solution

System components and technical architecture

The proposed solution is a fully integrated AI-Powered Real Estate Decision Intelligence Platform comprising eight interconnected system layers. Each layer addresses a specific gap identified in the AS-IS analysis and collectively delivers the TO-BE target operating model.

#	Component	Description
1	Centralised PostgreSQL Database	Primary relational data warehouse. Houses all cleaned and validated operational data with analytical views, indexes and role-based access controls.
2	Automated Data Ingestion (n8n)	Workflow automation platform that orchestrates scheduled ETL runs, validates incoming data, and triggers KPI alerts when thresholds are breached.
3	SQL-Based Analytical Models	Views and stored procedures that compute KPIs, calculate business metrics, and surface aggregated data for BI consumption.
4	Data Quality Validation Framework	Rule-based quality checks applied at ingestion to enforce completeness, accuracy and consistency standards across all data sources.
5	LLM-Powered Insight Generation Engine	API-connected large language model that analyses trends, generates investment recommendations, and produces narrative market commentary.
6	AI Recommendation System	Structured recommendation engine built on top of the LLM layer, targeting specific business decisions: acquisitions, pricing, and risk.
7	Power BI Executive Dashboards	Four interactive dashboards (Executive, Sales, Marketing, Customer Experience) connected live to the PostgreSQL warehouse.
8	RAG Knowledge Layer	Retrieval-Augmented Generation system allowing users to query business data in natural language, with answers grounded in actual data.

09 Business Benefits

Value delivered by the proposed solution

REDUCED MANUAL REPORTING EFFORT

Automated pipelines eliminate analyst time spent on data preparation, freeing capacity for insight-driven work.

IMPROVED DATA QUALITY

Validation frameworks enforce standards at ingestion, reducing errors that currently propagate into reports.

FASTER BUSINESS DECISION-MAKING

Real-time dashboards and AI summaries compress time-to-insight from days to seconds.

AUTOMATED MARKET INTELLIGENCE

LLM analysis continuously monitors market trends and generates recommendations without human intervention.

INCREASED SALES & MARKETING VISIBILITY

Granular performance dashboards surface agent, campaign and regional insights previously unavailable.

AI-ASSISTED INVESTMENT RECOMMENDATIONS

The recommendation engine supports acquisition, pricing and risk decisions with data-grounded AI analysis.

IMPROVED EXECUTIVE REPORTING

Power BI dashboards replace manual slide decks, providing live data directly to senior leadership.

10 Functional Requirements

What the system must do

ID	Category	Priority	Requirement
FR-01	Data Ingestion	Must	System shall automatically ingest Excel and CSV source files on a defined schedule
FR-02	Data Validation	Must	System shall validate all ingested data against predefined quality rules before storage
FR-03	Data Storage	Must	All cleaned data shall be persisted in a PostgreSQL relational database
FR-04	KPI Calculation	Must	System shall automatically compute all defined business KPIs from stored data
FR-05	KPI Alerts	Must	n8n shall trigger automated notifications when KPI thresholds are breached
FR-06	LLM Integration	Must	System shall connect to a LLM API to generate insights and recommendations
FR-07	NL Query Engine	Should	Users shall be able to query business data using natural language via the RAG layer
FR-08	Power BI Dashboards	Must	System shall provide four real-time Power BI dashboards for key business domains
FR-09	AI Recommendations	Must	System shall generate structured investment and risk recommendations via AI
FR-10	Report Automation	Should	Scheduled reports shall be automatically generated and distributed to stakeholders
FR-11	Role-Based Access	Must	Database and dashboards shall enforce role-based access controls by user type
FR-12	Audit Logging	Should	All data ingestion and transformation events shall be logged with timestamps

11 Non-Functional Requirements

System quality and performance standards

ID	Category	Requirement
NFR-01	Performance	Dashboard queries shall return results within 5 seconds under normal load
NFR-02	Availability	Platform shall maintain 99.5% uptime during business hours (07:00–22:00)

NFR-03	Scalability	System shall support data volume growth of 10x current baseline without redesign
NFR-04	Security	All data at rest and in transit shall be encrypted using AES-256 / TLS 1.3
NFR-05	Maintainability	All ETL scripts and workflows shall be documented and version-controlled in Git
NFR-06	Reliability	Failed ingestion jobs shall trigger automated retry logic and alert notifications
NFR-07	Usability	Power BI dashboards shall require no technical training for executive users
NFR-08	Compliance	Data handling shall comply with applicable data protection and privacy regulations
NFR-09	Recoverability	System shall support point-in-time PostgreSQL recovery within a 4-hour RTO
NFR-10	Auditability	All AI-generated recommendations shall include confidence scores and data sources

12 Assumptions & Constraints

Agreed conditions and known project limitations

Project Assumptions

- Historical data is available, accessible and suitable for initial platform load
- Business users will actively participate in KPI definition workshops
- PostgreSQL is approved as the primary database technology
- n8n will be used as the workflow automation platform
- Power BI licensing is available for all intended dashboard consumers
- LLM API services are accessible and API costs are within approved budget
- Source data owners will provide timely access to all required datasets
- Stakeholders will be available for UAT review within agreed timelines
- The IT team will provide the required infrastructure within project timescales

Project Constraints

ID	Category	Constraint Description
C-01	Data Quality	Initial dataset quality issues may impact analytical accuracy during early phases
C-02	AI Validation	All AI-generated recommendations require human business validation before action
C-03	API Costs	External LLM API usage will incur ongoing costs subject to budget approval
C-04	Data Accuracy	Platform output accuracy is directly dependent on source system data quality
C-05	Resource Capacity	Delivery timeline is subject to availability of the Data Engineering team
C-06	Infrastructure	Production environment must be provisioned before integration testing begins

13 Project Success Criteria

How we will know the project has succeeded

The project will be considered successfully delivered when all of the following success criteria have been verified, tested and formally accepted by the project sponsor and nominated stakeholder representatives.

Success Criterion	Priority	Acceptance Test
Automated Data Processing	Must	Business data can be automatically ingested, validated and processed without manual intervention
Real-Time KPI Visibility	Must	All defined KPIs are visible in Power BI dashboards within a 5-second refresh cycle

AI Insight Generation	Must	LLM engine successfully generates market insights and investment recommendations from live data
Stakeholder UAT Sign-Off	Must	All four Power BI dashboards pass user acceptance testing with nominated stakeholders
Automated Alerting Active	Must	n8n KPI alert workflows are tested, documented and operational in production
NL Query Capability	Should	RAG knowledge layer successfully responds to natural language queries with accurate data-grounded answers
Documentation Complete	Must	Full technical and user documentation delivered and accepted by the IT team
Performance Standards Met	Must	All NFR performance benchmarks validated through load testing

DOCUMENT SIGN-OFF

Approval & Authorisation

Role	Name	Signature	Date	Status
Project Sponsor / CEO				Pending
Head of Sales				Pending
Marketing Director				Pending
Data Engineering Lead				Pending
Business Analyst				Pending